Rutgers QEC & Beyond

SEASON 4 / TALK 3 FALL 2026 / ONLINE

#40 SERIES TALK

GeneCS: Synthesizing Resource-Efficient Code Surgery for Arbitrary Quantum Stabilizer Codes

MONDAY, NOVEMBER 09, 2026

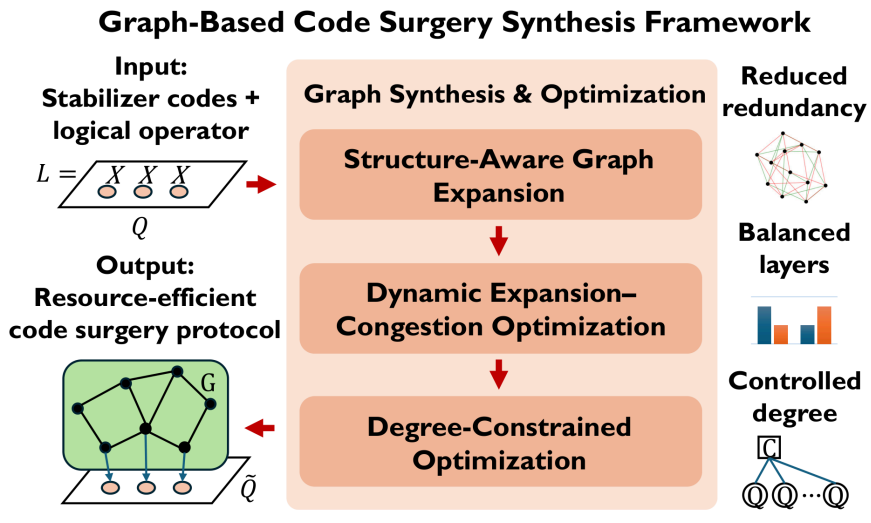
1:00-2:00 PM EST / 12:00-1:00 PM CST

ABSTRACT

Fault-tolerant quantum computers may need to combine different quantum error-correcting codes to achieve both efficient storage and computation. But how can we perform logical operations across arbitrary stabilizer codes without requiring a large number of additional qubits? Code surgery provides a general framework for joint logical measurements, yet existing constructions often incur substantial resource overhead.

This talk presents GeneCS, a compiler that treats code surgery as a graph synthesis and optimization problem. GeneCS reuses connectivity already present in the code, balances graph expansion against congestion, and incorporates degree constraints to construct compact ancilla systems while preserving code distance. Across single-code and cross-code benchmarks, it reduces the number of ancillary qubits and checks by roughly 85% compared with prior general constructions, while maintaining comparable logical error rates in simulation. I will also discuss how GeneCS can fit into a broader compilation workflow for heterogeneous quantum error correction.

GeneCS / CODE SURGERY SYNTHESIS



Join on Zoom

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ABOUT THE SPEAKER



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Junyu Zhou is a PhD student in the Department of Computer and Information Science at the University of Pennsylvania, where he works with Gushu Li in the Penn Quantum System Lab. He received his B.S. in Mathematics and Applied Mathematics from Wuhan University in 2024. His research focuses on software for fault-tolerant quantum computing, including quantum error-correction decoding, logical operations, and compilation. His work includes FusionQ, a real-time decoder with optimality guarantees; GeneCS, a compiler for fault-tolerant logical operations; and TopoLS, a surface-code compiler integrated into the TQEC ecosystem.

Supported by FINQ